

JuHyeong (Jenny) Seon

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PROFILE

Aerospace Engineering undergraduate with hands-on experience in CubeSat development, specializing in communication and ADCS subsystems. Passionate about pursuing a PhD in astrodynamics, with a research focus on trajectory optimization in the cislunar domain and space domain awareness (SDA).

EDUCATION

Saint Louis University

August 2023 - May 2027

Bachelor of Science in Aerospace Engineering | activities : Tau Beta Pi - Missouri Epsilon

Cumulative GPA: 3.83/4.0

EXPERIENCE

Space Systems Research Lab - SLU

August 2024 - Present

Communication Lead (August 2025 - May 2026)

- Led the communication subsystem team for CubeSat missions under NASA's CubeSat Launch Initiative (CSLI) and the University Nanosatellite Program (UNP), an AFRL-sponsored program, coordinating subsystem development, testing, and integration.
- Led troubleshooting of the UHF ground station, diagnosing hardware and software issues to improve carrier signal clarity and communication reliability.
- Conducted trade studies for S-band antenna and rotator systems, evaluating performance, cost, compatibility, and integration constraints to support system-level hardware selection.
- Designed ground station architecture and performed STK-based link budget analyses to evaluate communication performance and link reliability under mission-specific operating conditions.
- Coordinated with team members across communication, ADCS, and other subsystems to identify interface issues, establish testing priorities, and drive technical tasks toward mission objectives.

Attitude Determination and Control Lead (May 2026 - Present)

- Lead development of a MATLAB/Simulink ADCS simulation for CubeSat attitude dynamics and pointing using three-axis magnetorquers as the primary attitude actuators.
- Develop and evaluate magnetic attitude control and pointing performance through Freeflyer and Simulink, analyzing spacecraft attitude response and actuator behavior under different operating conditions.
- Prototype and test three-axis magnetorquers, using MATLAB-based analysis to evaluate coil parameters and actuator performance.
- Lead ongoing development and testing of a GNSS receiver for future integration with the spacecraft navigation and ADCS system.

Lab Manager (August 2025 - Present)

- Managed internal event organization, ensuring smooth execution of lab events at Saint Louis University.
- Facilitated general body meetings to promote collaboration and communication among members.
- Coordinated with other executive boards to enhance the effectiveness of lab initiatives.

Baleen Lab Research - SLU

Research Assistant (January 2025 - May 2026)

- Conducted unsteady CFD analysis in SC/Tetra to evaluate hydrodynamic performance of an autonomous surface platform designed for microplastic recovery in river and stream environments
- Developed and prepared 3D computational models and ~8 million-element meshes, applying local mesh refinement near the bow, filter housing, support structures, and wake regions to improve resolution of critical flow features
- Compared CFD results with Bernoulli-based theoretical predictions to assess physical consistency and validate simulation results.

SKILL

- Technical skills : Solidworks, SC/Tetra, MATLAB, STK1&2 Certification, Digital Twin (Sedaro), KiCad, Simulink, FreeFlyer
- Soft skills : Collaboration, Communication, Problem-Solving, Time Management, Troubleshooting
- Languages : English (Fluent), Korean (native)